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# Innovating Worker Safety: Designing Advanced Safety Helmets for Enhanced Heat Protection and Health Monitoring in High-Risk Industries

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## Abstract

In light of the rising temperatures worldwide and surging heat-related injuries on the job, innovating the Personal Protective Equipment (PPE) for high-risk industries can be vital. This paper aims to innovate worker safety by designing advanced safety helmets equipped with heat protection and health monitoring systems. It presents an innovative idea for a modified design of Safety Helmets, incorporating cooling and pulse-monitoring features to enhance the worker's safety and productivity. Using mechanical and electrical engineering principles, the design will incorporate Thermoelectric Coolers (TEC) and optical heart rate sensors for optimizing head temperature and monitoring pulse-rates. The design includes a two-layer Safety Helmet with a classic ABS outer shell and a polystyrene inner shell for superior thermal insulation. Powered by lithium batteries, the TEC device makes use of the Peltier phenomenon to establish a pleasant head temperature whereas the optical heart sensor can provide real-life data for the field managers, alerting them of any danger.

Preliminary results indicate that the proposed safety helmet can reduce head temperature, thus potentially decreasing heat-related health risks. The economic feasibility analysis suggests that the forecasted increase in productivity and safety monitoring justifies the additional costs of the design. Strict safety testing and adherence to international standards can ensure the viability of the Advanced Safety Helmet in high-risk industries such as construction, mining, and Oil & Gas. This paper introduces a new application of thermoelectric cooling and real-time health monitoring which, to our knowledge, has not been extensively implemented in the industry of Petroleum engineering. This paper not only ensures development in occupational health and safety measures, but it also lays a foundation for future integrations of solar power and AI-driven health-risk analysis.

## Introduction

Across any wide viewed industry, the safety of their employees is their priority. Health, Safety and Environment (HSE) is the vital department which follows protocols to ensure the safety of the workers. With an alarming rate of Global Warming and world record temperatures recorded in 2023, the rate of injuries due to heat waves has surged. Heat is the leading cause of death from all the weather-related conditions in the United States. Generally, helmets are often described as heavy, hot, and poorly ventilated, making them uncomfortable for long periods, especially in warm environments. Workers frequently cite discomfort, excessive warmth, and poor fit as major deterrents. This mostly causes them to avoid wearing the helmet during work shifts, which is not a safe act. According to the same source of the Occupational Safety and Health Administration, an average of 3,389 injuries occurs yearly in the USA due to heat-related illness. Not only that heat strokes and nausea can be a symptom, but heat also aggravates existing conditions such as asthma, heart diseases and kidney failure. Another disturbing allegation is that 6500 workers died in the construction of the World Cup stadiums in Qatar, mainly because of the overworked workers in the extreme heat.

The International Labour Organization (ILO) reports that excessive heat causes approximately 22.85 million occupational injuries and 18,970 fatalities annually. Adding on, jobs which are carried out in open-air environments, such as in the Construction, Mining or Oil & Gas industries require a direct exposure to the sharp sun as well as wearing the Personal Protective Equipment (PPE). This uniform usually and generally consists of body protection, thick footwear, hand gloves and a safety helmet. Therefore, workers often must fight both the heat of the sun and the over insulation of the PPE. Due to its ability to save lives, the PPE cannot be removed or taken lightly. Nonetheless, alterations to it can ease the work in the hot weather. Throughout this paper, the adjustment of the safety helmet will be looked at closely and its feasibility will be discussed.

The Safety Helmet is one of the most crucial PPE components which cannot be ignored or forgotten. Other than saving the lives of engineers who are under the risk of falling objects, it also reflects the sun rays in order to prevent heat strokes. While they are crucial for protection, Safety Helmets can trap heat around the head, further leading to illness. Therefore, a helmet can be designed for the harsh conditions, such as the extreme temperatures of the Middle East reaching 40 Degrees Celsius. For that reason, during this paper, a Safety Helmet will be discussed having two new features: a cooling feature and a monitoring system for the workers' safeties.

## Literature Review

Personal Protective Equipment (PPE) has been a life-saving tool for decades. The PPE has evolved significantly with recent advances in technology, especially in the fields of monitoring the environment around the user and their physical and psychological conditions. These features are highly useful and beneficial in high-risk industries such as Construction or Oil & Gas. In the Construction industry, DAQRI Smart Helmet was one of the first companies to implement augmented reality and real-time sensor feedback in order to help workers imagine and see the risks of the field and to learn how to be more efficient without being in the line of fire. This advancement exited the market and pushed other companies to start thinking about innovations in the PPE. In the industry of firefighting, some helmets were designed with thermal sensors, heads-up displays and communications systems in order to minimize the risk of going into a hazardous environment blind. Moreover, there is also an integration of low-power wireless modules like the ESP32, which allow health features monitoring of the user with live data, without creating a significant change in the helmet's ergonomics.

More technological advances were observed and created by Kenzen and Hexoskin in the Construction industry, where smart vests use biometric sensors to evaluate the worker's physiological state by live transmission to mobile apps. These vests have integrated biometric sensors that continuously monitor crucial health indicators such as: heart rate, body temperature, respiration rate and hydration status. After taking these readings, the data is transmitted to mobile applications, alerting the supervisors of any health risks. This innovation demonstrated positive results, including a reduction in heat-related incidents and improved worker productivity.

Another notable innovation is the Gatorade Gx Sweat Patch, which integrates biosensor technology in occupational health and safety. This patch uses a sensor to test the composition of the sweat on the worker's skin, providing data on fluid and electrolytes loss through a mobile application. The prevention of dehydration in demanding industries and hot environments is difficult when the PPE traps body heat and sweat under the uniform. Being well-hydrated will improve endurance, reduce downtime and prevent heat-related accidents. Therefore, these patches will ensure that workers' safety is always a priority by ensuring their hydration.

## Focus on Safety Helmets: Current Limitations and Opportunities

The Safety Helmet is one of the most crucial PPE components which cannot be ignored or forgotten. Other than saving the lives of engineers who are under the risk of falling objects, it also reflects the sun rays in order to prevent heat strokes. While they are crucial for protection, Safety Helmets can trap heat around the head, further leading to illness. Therefore, a helmet can be designed for the harsh conditions, such as the extreme temperatures of the Middle East reaching 40 Degrees Celsius. For that reason, during this paper, a Safety Helmet will be discussed having two new features: a cooling feature and a monitoring system for the workers' safeties.

## Innovative Helmet Features and Technical Design

### Part 1. Cooling

Regulating head temperatures and lowering the ambient temperature using this new feature can lead to numerous added benefits for the workers. According to a study carried out in India on 70,000 plants/ factories, the productivity of a worker drops by as much as four percent per every Degree Celsius above 27 Degrees Celsius. That being so, a cooling system could be highly useful for employees working in extreme weather conditions by increasing their productivity and decreasing health risks. Moreover, workers in these conditions have to be monitored for health hazards. As a result, a monitoring system will also be installed which directly transmits signals to the manager of the site in case of irregularities in the pulse rates of the workers.

Designing this Safety Helmet requires a deep mechanical and electrical understanding. The cooling system will first be discussed. In this case, the unit will consist of two layers. The outer layer is the resistant and hard shell. Most helmet shells are made of ABS (Acrylonitrile Butadiene Styrene), a hard, shock absorbing material. Figure 1 below shows the yellow material of the helmet illustrating the typical ABS material with which it will be constructed. On the other hand, the inner layer will be a lighter component which is a good thermal insulator, such as polystyrene. The cooling system will be an active battery-powered system, made of thermoelectric coolers (TECs), which are also known as Peltier devices. The TEC is made of 3 main components. Firstly, it has two different conducting materials called Thermoelectric Couples. The second component is the Heat sinks, which absorb the hot conductor and release the heat out of the helmet, as seen on the sides of the helmet design below. Lastly, it has a DC power supply, which is usually lithium batteries as seen in the back of the helmet in Figure 1 below. The Peltier effect happens when current runs through two different conductors, where heat gets absorbed on one side and released on the other, creating a temperature difference. When a current flows through it, a cooling action occurs that lowers the temperature of the helmet. This cooling section will be in the interior of the helmet whereas the hotter side will be in the outer layer of the helmet. The rechargeable batteries will be placed in the rear-end of the helmet, secured safely. Lastly, there will be a digital temperature sensor on the lining of the helmet, such as the DS18S20+, which will be connected to a microcontroller that activates the TEC device upon exceeding the 27 Degrees Celsius mark.

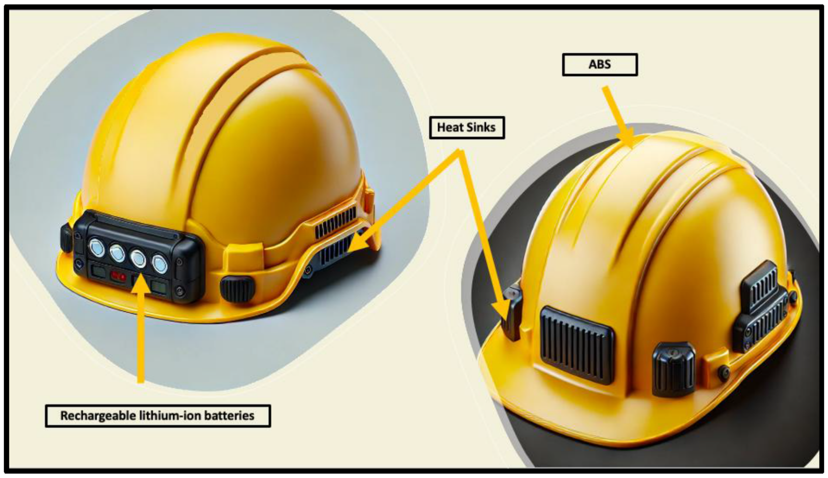


Figure 1—Techical Design of Safety Helmet Cooling System

## Part 2. Pulse Monitoring

Secondly, part two of this design is the pulse monitoring. Doing so, a pulse sensor is required. Most commonly, an Optical heart rate sensor can be used to measure the person's heart rate by shining bright light through the skin and monitoring the changes in blood flow by the reflection of the light back. The device used to read the reflected light is called a photodetector. This Optical heart rate sensor could be placed around the temple, making sure that it remains secure and does not cause discomfort. Again, the power source of this device can be a small lithium battery that can be recharged. Similar to the designed cooling system, a microcontroller should be used for the wireless communication capabilities. For cost-effective reasons, the ESP32 chip can be utilized, making use of its Wi-Fi and Bluetooth features. This device is known for its energy efficiency and wide range of documentation, offering real-time data support. As seen in Figure 2 below, the pulse-monitoring system will be placed on the side of the safety helmet, giving no discomfort to the user.


Figure 2—Technical Design of Safety Helmet Pulse Monitoring System

## Economic Feasibility

Economic feasibility is a primary factor in determining the implementation of a project. Standard simple safety helmets can start from $15 and can reach prices above $200. These average costs include the inner and outer shells; it is an essential cost for any industry contractor if they want to have workers on-site. The first additional component, which is the TEC, varies its price from $3 to 15$ depending on the size and efficiency. Lithium-Ion batteries can vary widely in prices, from $5 up to $50 depending on the quality of these batteries. According to a study carried out by BloombergNEF, strategic research covering technologies and global commodity commodities, the average Lithium-Ion battery costs $151 per kWh. If it is assumed that the Peltier device consumes 5 Watts, and that a working day is 8-hours, then the price of the battery will be rounded up to $7. The wiring of the whole system, as well as the connectors, can average a price of $5 dollars. Moreover, the digital temperature sensor, DS18S20+, can cost around $7. In the case of the pulse monitoring, there are two main components; the Heart Rate sensor and the ESP microcontroller and their average prices are both $5. Therefore, the total added cost of the whole system, taking average prices, would be around $40. Certainly, when purchased in bulk, their prices will be much lower than the estimated added cost. Despite the significant additional cost of the helmet relative to its original price, the potential increase in the productivity and the vital monitoring of the employee's health justifies this investment. An average summarized total cost is shown in the table below.

**Table 1—Average total additional cost summary**

| Component | Average Cost ($) |
|---|---|
| TEC (Thermoelectric Cooler) | $3 - $15 |
| Lithium-Ion Battery (8-hour runtime) | $7 |
| Wiring & Connectors | $5 |
| Temperature Sensor (DS18S20+) | $7 |
| Heart Rate Sensor | $5 |
| ESP Microcontroller | $5 |
| **Total Additional Cost** | **~$40** |

When looking at the extra costs, it could seem unattractive for companies that purchase the safety helmets. Nevertheless, the potential fines and costs that could be imposed on employers for potential employee injuries is much higher. For example, in 2018, a carrier in Los Angeles suffered from hyperthermia in 117F heat. This led to a $149,664 in fines proposed by OSHA for recordkeeping and General Duty Clause violations. Additionally, a Merced Farm Labor Contractor was fined $262,700 in July 2008 by Cal/OSHA after 17-year-old Maria Isabel Vasquez Jimenez died of heat stroke during work. In the last 10 years, an estimate of several million dollars were given, with notable fines of around $200,000 per case. Therefore, with that being stated, the additional cost of the innovative Cooling Safety Helmet is minimal compared to the fines that could be paid even for minor heat-related injuries.

## Environmental Impact and Sustainability

When designing safety helmets, the first function that comes to mind is its immediate effect, which is to protect people from impact. It is important to notice that every product that is created in the massive oil & gas industry can have a ripple effect on the planet. Which is why it is a responsibility to consider the environmental impact of the helmet material on the planet. The ABS and Polystyrene materials used in the design were maintained from the traditional helmets despite being derived from fossil fuels, as both can be processed through established recycling streams. This means that the lifestyles of these helmets could be extended, and their impact could be minimized. Secondly, the introduction of the TECs and the Pulse Monitoring devices will require batteries. In this case, Lithium-ion rechargeable batteries were picked in order to minimize disposable battery waste, but careful end-of-life disposal should be carried out.

## Safety Tests Required

Besides considering the costs, the product must go through safety testing and certifications. The International Safety Equipment Association (ISEA) is a trade association for companies that provides leadership and expertise in product standards regarding safety and in compliance with the American National Standards Institute rules. Not only that, but rigorous testing has to be done to check if there are any side effects of wearing this helmet and examining any possible risks. Thermal Performance Testing will be required to verify the effectiveness of the heating system as well as ensuring that the head temperature is within range to maintain the safety of the worker. Moreover, the electrical hazards are also possible and need to be considered in these harsh conditions, therefore, an electrical testing will also be undergone. To add on, ergonomics and usability trials are also crucial testing points as the safety helmet has to be convenient to the user, not causing any discomfort or issues. The table below gives a summary on the safety and performance standard testing.

| Aspect | Standards and Tests | Description |
|---|---|---|
| Impact/Penetration Safety | ANSI/ISEA Z891<br>EN 397 | For Impact Protection and Mechanical Hazards |
| Thermoelectric Cooling System | Performance Testing<br>IEC 60950-1<br>IEC 60529 | For Efficiency of cooling, electrical safety and durability |
| Electrical Safety | IEC 62133 (Lithium-ion battery safety) | For ensuring the battery's safety for electrical hazards |
| Sensor Accuracy for Pulse Monitoring | IEC 60601-1-2<br>ISO 10993 | For Sensor accuracy in its readings and to prevent skin irritations |
| Wireless Communication & EMC | EN 50566, EN 50360, EN 62209 | For ensuring that the wireless connection does not affect any other device and does not affect the user |
| Software & Cybersecurity | ANSI/CAN/UL 2900 | For user protected cyber threats |
| Ergonomics/Usability | IEC 60601-1-6 | For ensuring that the helmet is comfortable |
| Environmental & Durability Tests | IEC 60068 | To ensure that the tests are reliable and well-done under harsh conditions |

## Conclusion

In summary, this paper presents a pioneering approach to enhancing occupational health and safety through the development of an advanced safety helmet tailored for high-risk industries. By integrating thermoelectric cooling technology and real-time health monitoring systems into a robust, ergonomically designed helmet, the proposed solution directly addresses the escalating risks posed by rising global temperatures and heat-related workplace injuries. This innovative approach in the physical industries warrants further in-depth research to assess its feasibility and applicability. This research mainly investigates an improvement in the working conditions in the hot weathers as well as an increase in productivity. The modified Safety Helmet creates an added cost of around $40 based on this short research, nonetheless, it can be highly beneficial in the long-term. The increase in productivity can further lead to higher profits that cancel out the added costs. This is considered a Capital Cost, only paid once at the beginning of the project. Moreover, multiple firms violated the OSHA standards of working in the heat, leading to very high fines which would not have been the case if the workers did not feel this extreme heat. Lastly, and most importantly, the health of an employee should always be put first and designing this helmet can be the first step to preventing more heat incidents. Therefore, integrating advanced technologies and real-time health monitoring, such as the modified Safety Helmet discussed in this paper, into standard PPE represents a significant leap forward towards the initiative of occupational health and safety.

## Further Recommendations

Given that one day this modified Safety Helmet is implemented, more adjustments could be made. Looking into the future and given that Gulf Countries are high producers of Oil & Gas as well as having high construction activity, a Solar Panel can be implemented in the helmet in order to save energy costs. Not only that, but with the development of Artificial Intelligence, machine learning algorithms could be implemented to analyze data and predict potential health risks and alert the employee. Furthermore, ongoing research and feedback from real life field applications can be crucial to the continuous improvement of equipment designs, ensuring the safety and improvement of workers. Ultimately, the adoption of such advanced PPE has the potential to significantly mitigate heat-related health risks, foster safer working environments, and drive innovation in personal protective equipment across various high-risk industries. Enhancing helmet comfort and improving thermal insulation may significantly increase user compliance, thereby reducing the incidence of accidents and heat-related fatigue in the workplace.

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